

Image-Free Tumor Segmentation of Soft Tissue Using a Minimally Invasive Robotic Palpation System

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Abstract—Tumor segmentation is crucial for surgical planning and precise tumor resection for effective treatment. Traditionally, tumor localization has been performed using medical imaging techniques such as CT and MRI or through direct palpation by surgeons. However, in minimally invasive robotic surgery (MIS), these methods have limitations, including registration errors with imaging and inaccuracies caused by the subjectivity of palpation by surgeons. In this study, we introduce a robotic palpation system and an image-free process for MIS tumor segmentation using a robot. Our proposed system enables precise tumor shape differentiation through direct robotic palpation. For this, the robotic palpation system collects surface shape information through the proposed process, allowing tissue palpation at specific depths according to surface curvature. Additionally, it visualizes stiffness maps, enabling image-free tumor segmentation. In experiments using this system, evaluation of planar and curved phantom models demonstrates precise segmentation at targeted sites, with sensitivities of 0.9634 and 0.9729, and specificities of 0.9646 and 0.9878, respectively. Validation on ex-vivo porcine liver models further confirms the efficacy of our approach.

Index Terms—Image-free, robotic palpation, robot-assisted minimally invasive surgery (RMIS), surface reconstruction, tumor segmentation.

I. INTRODUCTION

TUMOR segmentation is the process of identifying the regions where tumors are present within tissues and distinguishing tumor regions from normal tissues. This process is essential for accurate diagnosis and surgical planning, particularly in tumor resection procedures [1], [2], [3]. In general,

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image-based methods utilizing medical images such as CT, MRI, and PET and palpation-based methods are used for tumor segmentation.

Medical imaging is a widely used technique to assist in surgical planning by visualizing internal organs and identifying tumors within them [4], [5]. Through the image processing, these images help distinguish tumor regions and reconstruct the 3D shapes of organs and tumors. However, image-based tumor segmentation has several limitations. One major drawback is the significant time and cost required for medical image acquisition. The process typically takes between 10 to 50 minutes, increasing the likelihood of patient movement and associated errors [6]. Additionally, discrepancies can arise when aligning medical images obtained through different modalities, such as CT and MRI. Furthermore, due to the nature of soft tissue deformation and movement during surgery, the actual intraoperative tumor location may differ from the segmented tumor location in medical imaging.

Additionally, palpation has been utilized to enhance the accuracy of tumor segmentation during tumor resection surgery [7], [8]. This technique involves the physician's tactile assessment of organ properties, helping to identify abnormal tissue characteristics such as size, location, and shape - key factors for disease diagnosis and surgical planning. Palpation plays a crucial role in ensuring effective and safe surgical procedures, particularly for radical tumor removal [9]. However, its reliance on physician's subjective assessment presents challenges to accuracy [10].

Recently, minimally invasive surgery (MIS) has been recognized as a suitable approach for various tumor resection procedures [11], [12], [13], [14]. MIS offers several advantages, including shorter recovery times, reduced patient discomfort, and minimal scarring [15], [16], [17]. Due to these benefits, MIS is becoming increasingly prevalent in clinical practice, and research focused on improving tumor segmentation accuracy in MIS is actively ongoing [18], [19]. However, this approach introduces challenges such as friction at the trocar port and restrictions on device movement, which may compromise palpation accuracy. Integrating force sensors into MIS instruments enables a quantitative assessment of organ properties by measuring force values during contact. However, the clinical adoption of force-sensing capabilities remains limited [20].

To overcome these challenges, recent research has actively explored palpation-based tumor segmentation using robots [21],

[22], [23], [24]. Robotic palpation has the advantage of providing more objective and quantitative assessments compared to conventional palpation methods. However, existing robotic palpation-based tumor segmentation studies also have limitations. One major challenge is that most of existing robotic palpation requires prior acquisition of the organ's surface shape, which typically relies on medical imaging or cameras. However, the use of medical images may lead to the previously mentioned issues related to time and cost. Moreover, in robot-assisted MIS (RMIS) environments, reconstructing surface geometry using cameras is highly challenging due to difficulties in obtaining high-quality images and aligning them accurately. In addition, some existing studies have attempted to detect tumors using specially designed sensors or deep learning-based palpation data analysis without the use of medical imaging [25], [26], [27], [28], [29]. However, most of these methods are limited to flat tissue surfaces and are still limited in terms of applicability and generality in real curved tissues or MIS environments.

To overcome the limitations of previous studies, this study proposes a robotic palpation system capable of tumor segmentation without relying on medical imaging or cameras. Unlike existing methods that require prior medical image acquisition or surface reconstruction using external sensors, the proposed method introduces a novel approach in which a single robotic system directly obtains the tissue surface information and performs objective palpation to generate a stiffness map. This enables precise tumor segmentation without the need for coordinate alignment. Additionally, the system allows for the objective measurement of tissue stiffness without prior knowledge of the tissue shape, facilitating accurate tumor segmentation in RMIS environments. The effectiveness of the proposed robotic palpation system is validated through phantom and ex vivo experiments, demonstrating its potential for clinical application.

II. METHODS

Fig. 1(a) illustrates the overall process flow for image-free segmentation using the proposed robotic palpation method. To perform the proposed robotic palpation, the first process, surface reconstruction, is performed through a master-slave tele-operation system. First, point cloud acquisition is performed by scanning the surface of the soft tissue with the slave robot's end effector to obtain the position $\mathbf{p}_{end\ effector}$ of the robot's end-effector. Next, using the obtained position information, the surface is reconstructed to generate the surface normal vector $\mathbf{N}$ and the reconstructed point $\mathbf{p}_{recon}$, thereby acquiring the surface geometry. Secondly, a robotic palpation process is performed. Using the $\mathbf{N}$ and $\mathbf{p}_{recon}$ obtained from the previous process, a probe path with a specific indentation depth is generated through MATLAB programming. The end-effector of the slave robot follows the generated probe path, applying pressure to the soft tissue during palpation. Lastly, the third process, Tumor segmentation, is performed. In the previous process, when the robot's end-effector presses the tissue, the 3-axis force is measured by the force-torque sensor attached to the robot's tool.

The measured force $\mathbf{F}$ and the corresponding $\mathbf{p}_{end\ effector}$ go through post-processing to estimate the stiffness value $E(\mathbf{p})$ of the tissue at each position along the probe path. The estimated stiffness is densely interpolated to create a stiffness map, and by setting a threshold based on the stiffness values, specific boundaries above a certain stiffness can be segmented to distinguish the tumor. All the aforementioned processes are performed by a single robotic system. Since the reference coordinate system is the same, there is no need for separate kinematic calibration or coordinate system registration. Therefore, image-free tumor segmentation could be accurately performed.

The concepts underlying each process are succinctly elucidated in Fig. 1(b). As illustrated in Fig. 1(b)(i), the probe is directly in contact with the tissue in an image-free method, performing the point cloud acquisition along the probe's movement to obtain the tissue surface. Additionally, Fig. 1(b)(ii) shows that when the robot's probe is indented in the normal direction of the tissue surface, the reaction force transmitted to the probe can be measured. Since the typical difference in stiffness between tumor and normal tissue, reaction forces are different from each tissue type. Hence, by correlating the reaction force with indentation depth, the stiffness of each tissue can be estimated, typically expressed as Young's modulus. Finally, Fig. 1(b)(iii) illustrates the process of segmenting the tumor by setting a threshold in the visualized stiffness map, using the stiffness difference between normal tissue and the tumor tissue.

A. Robotic Palpation System

In this study, the robotic palpation system consists of a master device, a slave device, and a palpation tool, illustrated in Fig. 2(a). First, the master device, named Touch by 3D Systems, is a haptic device with 6 degrees of freedom (DOF) and is controlled by the physician. Second, the slave device, named UR5e from Universal Robots, is a 6-DOF robot manipulator executing tasks on the patient's side under the manipulation of the physician. These devices operate in a tele-operated master-slave configuration, with control achieved at a loop speed of 2 milliseconds using real-time task scheduling facilitated by Linux 5.4.89 and Xenomai 3.2 kernel on Ubuntu 18.04.01 LTS OS. Both the master and slave devices are fixed on an optical table alongside a phantom model. Finally, the palpation tool is attached to the end of the slave device, comprising a long rod, a 6-axis force/torque sensor named HEX12 by Resense, and a ball probe, as depicted in Fig. 2(b). Here, the elongated rod of the tool is constructed from SUS340 material, measuring approximately 210 mm in length. Its design is optimized to handle a wider area within the patient's body, specifically tailored for applications in minimally invasive surgery. Additionally, the force/torque sensor incorporated in the system exhibits a narrow force range coupled with high resolution, showcasing commendable performance characteristics suitable for the precise control of soft organs, in low-noise environments. Furthermore, a 2 mm diameter ball probe is attached to the end of the force/torque sensor which senses the reaction force from the tissue as the ball probe moves on the tissue surface and stores this data at a pace at the same

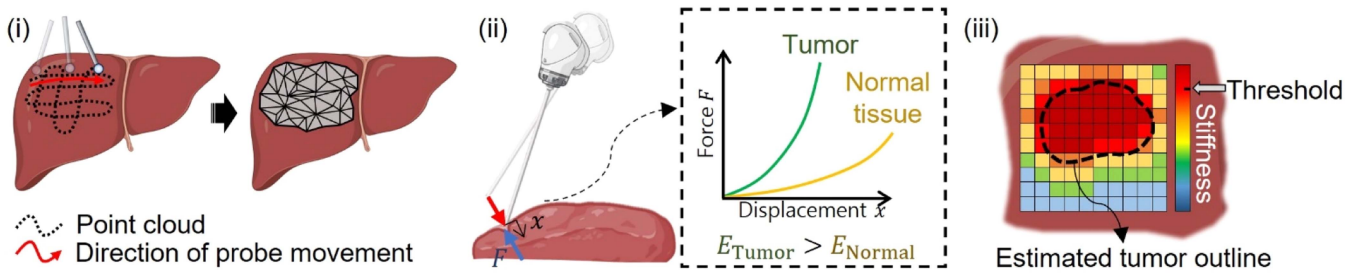
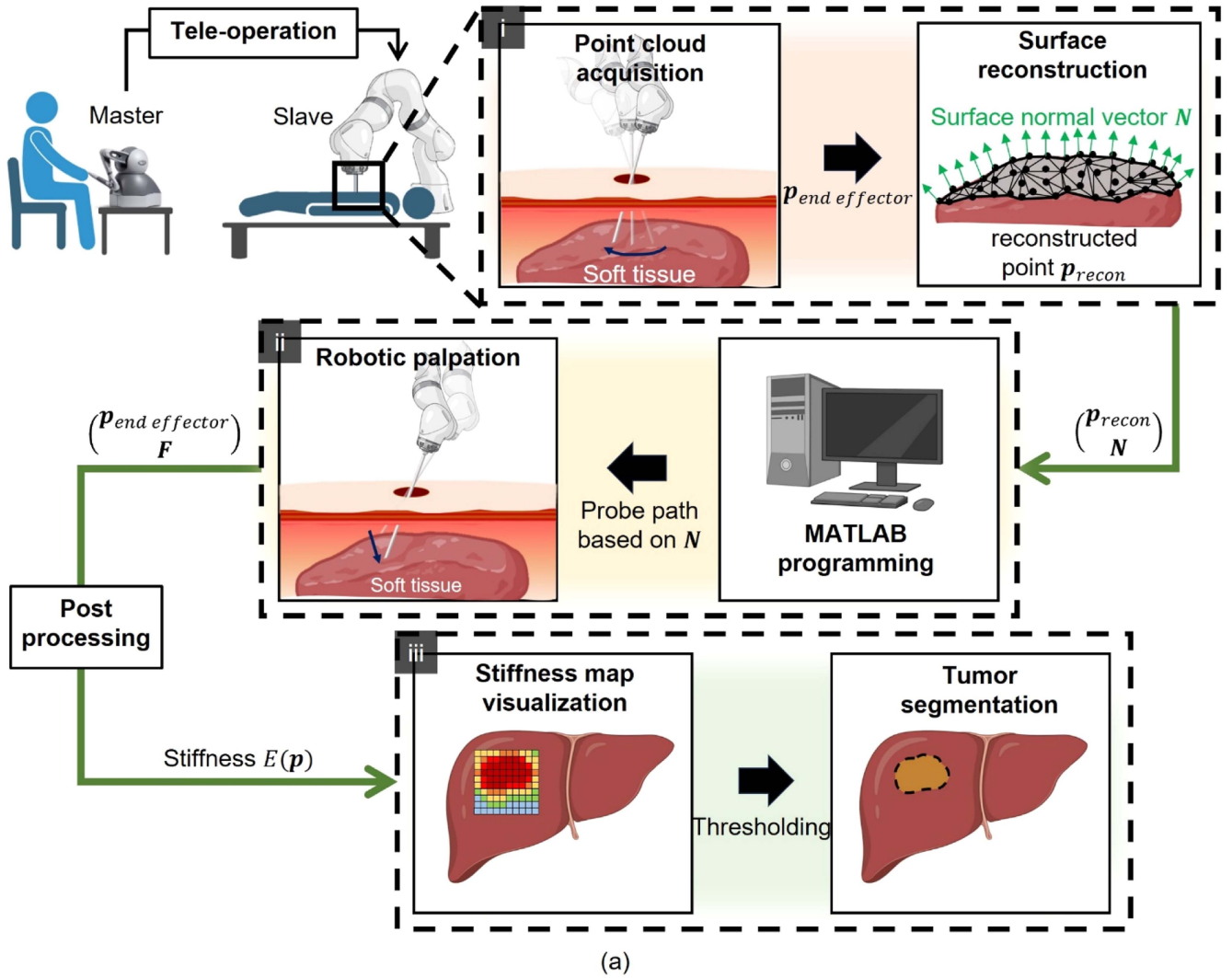


Fig. 1. (a) Overall procedural framework for image-free segmentation using the proposed robotic palpation technique and (b) concept underlying each process of the proposed method.

speed as the robot control loop speed. The sphere-shaped probe used not only ensures a consistent contact area during soft tissue probing but also serves to minimize tissue damage. Moreover, in estimating the precise point of contact between the palpation tool and the tissue surface, the end coordinates of the ball probe are utilized. It is worth noting that employing a smaller diameter ball probe, such as the 2mm variant employed in this setup, aids in reducing location errors, thereby enhancing the overall precision of the system.

B. Surface Reconstruction Process

Generally, the surface information of the targeted tissue is delineated by the positions and orientations of a collection of points constituting its surface. The initial point cloud generated by the proposed robotic palpation system represents partial and raw data, capturing only a portion of the tissue surface information. To achieve a comprehensive understanding of the tissue surface, it is imperative to undertake a reconstruction

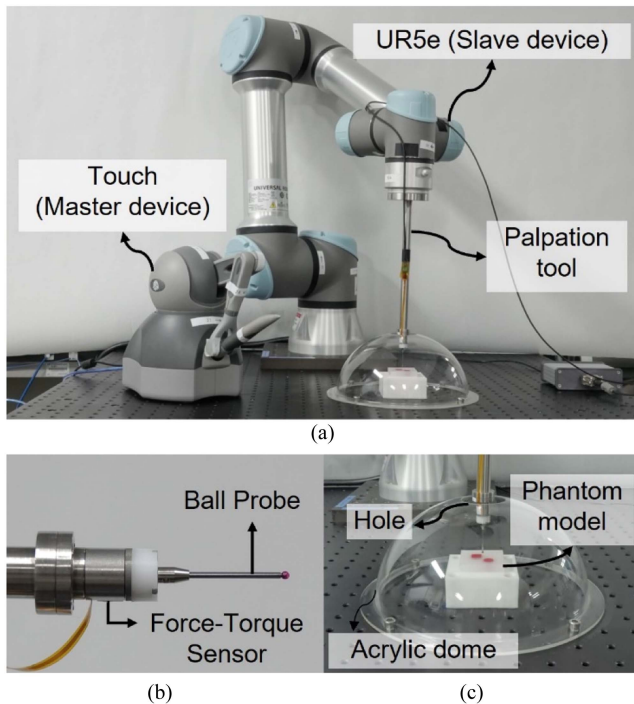


Fig. 2. (a) Robotic palpation system, (b) ball probe and force-torque sensor in palpation tool, and (c) phantom test setup.

process to extrapolate the complete surface information. Consequently, both point cloud acquisition and subsequent surface reconstruction procedures are indispensable for obtaining precise and comprehensive tissue surface data.

Initially, the proposed robotic palpation system performs point cloud acquisition to acquire the positional information of points on the surface of the organ. Throughout this process, when positional input is provided from the master side, the slave robot instantly tracks and follows these specified positions. As the ball probe situated at the end of the robot moves on(over) the tissue surface, it instantaneously records the positional information of the ball probe's end, thereby storing points in real-time. This collection of points is termed a point cloud. This study performed the point cloud acquisition process through the remote center of motion (RCM) constraint control and position-based impedance control, as described in our previous research [6]. First, RCM constraint control involves the remote control of the robot's rotational and translational motion relative to a fixed point in minimally invasive surgery, as shown on the left in Fig. 3(a)(i). At this time, the fixed point is set based on the hole in the human body for inserting surgical tools during minimally invasive surgery. This point is defined to as a virtual RCM point on the robot. RCM constraint control is a method used to control the position of the probe tip and RCM point by converting a total of five task control variables into five joint velocities using the alternative Jacobian [6]. With this control, the robot's tool moves constrained to the RCM point and follows the position input from the master. Second, position-based impedance control is to control the position of the slave according to the input to the master while controlling the slave to move in contact with the tissue without damaging it. Position-based impedance

control, as shown in the right figure of Fig. 3(a)(i), constructs a virtual damper and spring between the tissue and ball probe with constants M , C , K , and external force F_{ext} . This allows the actual slave to control the position x without damaging the tissue, even if the master side gives an input x_d to contact deeper than necessary into the tissue. With the two control techniques introduced, the raw data of the tissue surface information can be obtained in the form of a point cloud.

The point cloud obtained through the introduced point cloud acquisition is generated by the ball probe attached at the end of the slave robot directly contacting the tissue, so it does not contain information about a significant area that did not make contact. Therefore, a surface reconstruction process is required to reconstruct the point cloud using the Point2Mesh deep learning algorithm. The Point2Mesh deep learning algorithm is an unsupervised deep learning algorithm that takes as input a point cloud and a rough initial mesh [30]. This algorithm is robust to noise and has the feature of effectively supplementing missing data. To reconstruct the surface, the point cloud is down-sampled and used as input data and is trained as a uniform point cloud with an initial mesh generated based on it. Through this mesh-based learning, not only the position information of the points but also information of the surface normal vector is obtained. Finally, the reconstructed surface information of the patient area can be obtained image-free from the point cloud data, as shown in Fig. 3(a)(ii), after undergoing the smoothing procedure. Previous research reported that this surface reconstruction process has an accuracy of up to 0.35 mm [6].

C. Robotic Palpation Process

The proposed robotic palpation is the process of estimating the quantitative stiffness of the tissue surface by probing the tissue by a robot based on the reconstructed surface of the tissue obtained in the Section II-B. In this study, to probe the tissue uniformly and accurately, the probe path is generated by utilizing the tissue reconstructed surface information. Then, as the robot's probe tip moves along the generated path and probes the tissue, the force values measured from the force-torque sensor connected to the probe tip can be used to estimate the stiffness of the tissue along the probe path.

In the robotic palpation process, a probe path is created to ensure uniform probing of the entire tissue and a certain depth of probing in the surface's normal direction. First, to achieve uniform probing of the entire tissue, a regression line is generated for each sector at regular intervals on the entire surface, as shown in the left image of Fig. 3(b)(i), and points are extracted from the generated line at equally spaced intervals and set as probing points. Since the reconstructed surface of the tissue is composed of points arranged disorderly, a regression line is generated by applying the least square method to the points within one sector. Here, sector n is a specific range set along the y-axis. Once n regression lines are generated at regular intervals, points are uniformly extracted on all regression lines. After that, among these extracted points, the point closest to the points on the reconstructed surface is set as the probing point. Second, to maintain the same indentation depth at all set probing points,

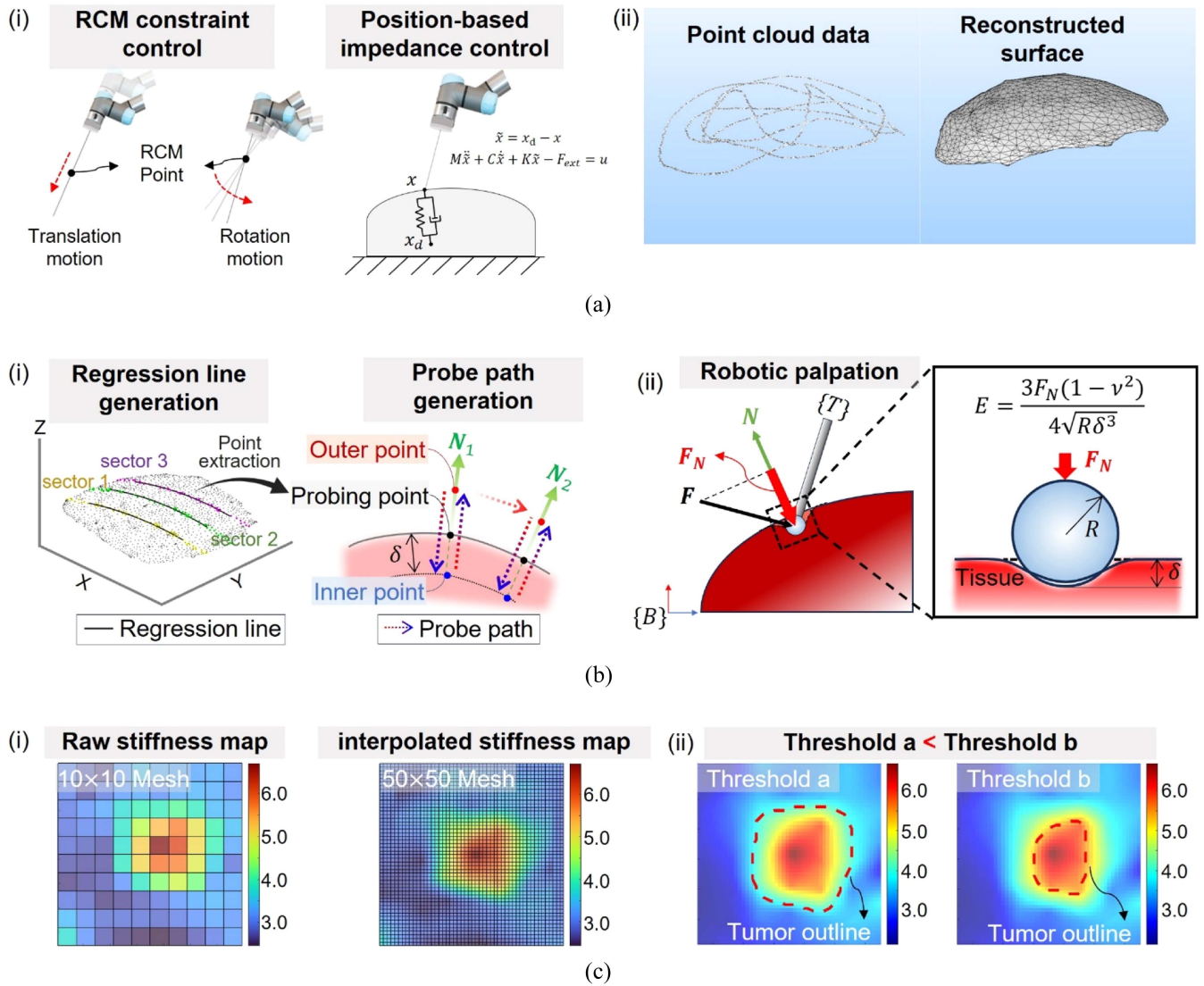


Fig. 3. (a) Surface reconstruction process, (b) robotic palpation process, and (c) tumor segmentation process.

the end point of the indentation is set at a certain depth in the internal direction of the tissue from the probing point using the surface normal vector N , as shown on the right in Fig. 3(b)(i). The end point of the indentation is defined as the inner point, and the point in the direction away from the tissue is defined as the outer point. For a single probing point, moving from the outer point to the inner point and then from the inner point to the outer point constitutes one indentation process. By repeating the same process for the next probing point, uniform robotic palpation can be performed across the entire tissue surface when applied to n sectors. Eventually, a final probe path can be generated for all probing points that move in the order of outer point, inner point, and outer point.

Subsequently, the force exerted by the palpation tool on the tissue is generated when the robot's probe tip follows the probe path generated by the robotic palpation process and indents the tissue in the surface's normal direction. During this process, a reaction force is exerted on the palpation tool by the action reaction, and this reaction force is sensed through the force

torque sensor. The mechanical properties of the tissue can be estimated from the force measured due to the contact of the ball probe with the tissue [18]. Specifically, this study aims to estimate the stiffness of a viscoelastic tissue from the contact between a spherical rigid ball probe and a viscoelastic tissue, considering the complex characteristics of the tissue. In particular, according to the Hertz solution, when a rigid sphere with radius R presses on a viscoelastic object, Young's modulus (E) of the tissue is expressed as (1) [31]. Under these conditions, the stiffness of the tissue is an absolute value rather than a relative one.

$$E = \frac{3F_N(1-\nu^2)}{4\sqrt{R}\delta^3}. \quad (1)$$

Here, ν represents Poisson's ratio, δ denotes indentation depth, R signifies the radius of the indenter, and F_N denotes the force projected from the sensing force onto the surface normal vector N . As shown in Fig. 3(b)(ii), N is a normal vector generated with respect to the base coordinate system $\{B\}$, but the force

F sensed by the force-torque sensor is a vector generated with respect to the coordinate system $\{T\}$ of the palpation tool where the sensor is located. Also, N and F are not parallel to each other due to friction forces, control errors, etc. This means that F contains other force components such as friction forces other than the indentation force, so only F_N , the projection of F in the N direction, is used for stiffness estimation. In the end, the stiffness of the tissue can be estimated using the force F_N in the direction normal to the surface of the tissue.

D. Tumor Segmentation Process

Through the robotic palpation process described in Section II-C, the position of the probing point and its corresponding stiffness value along the tissue's surface normal direction can be obtained. Using these stiffness values, a stiffness map of the target area is generated. However, since the probe position and stiffness values represent only a localized region of the target area, interpolation is required to estimate the stiffness values in areas beyond the actual probe points and construct a comprehensive stiffness map. With these interpolated stiffness values, a rough raw stiffness map, like the left image of Fig. 3(c)(i), can be created by generating a mesh. By interpolating more compactly, a dense processed stiffness map, like the right image of Fig. 3(c)(i), can be created, and through this, it is possible to ultimately visualize the stiffness information of the tissue.

On the other hand, it is generally known that tissues with tumors have higher stiffness compared to normal tissues [32]. Therefore, in the stiffness map, normal and tumor tissues can be distinguished visually through the color map. In this study, we aim to separate the stiffness area above a certain stiffness value among the interpolated stiffness values from the stiffness map and define its boundary as the outline of the tumor. This certain stiffness value represents the threshold for the tumor stiffness. As shown in Fig. 3(c)(ii), by adjusting the threshold, we could distinguish a wider tumor area when the threshold is small, and a narrower tumor area when the threshold is large. Here, the threshold can be determined according to the stiffness of normal tissue. In general, it is known that tissues with tumors have a stiffness that is about 2-3 times higher than normal tissue [33]. Therefore, a threshold can be set as a multiple of the stiffness of the area that is judged to be normal tissue.

E. Performance Evaluation (Phantom Test and Ex-Vivo Test)

To assess the efficacy of the proposed robotic palpation system, two types of tests, namely phantom and ex-vivo tests, are conducted. All test was conducted utilizing a robotic palpation system to simulate a minimally invasive surgery (MIS) scenario, devoid of surface information on the models. The procedure encompassed point cloud acquisition for surface reconstruction, robotic palpation guided by probe path generation for assessing tissue stiffness, and tumor segmentation utilizing the stiffness map generated from the tissue. As depicted in Fig. 2(c), the tests are executed utilizing an acrylic dome that encloses the model. The hole situated atop the acrylic dome simulates the entry point of the human body for the insertion of surgical instruments

during minimally invasive surgeries. Throughout all stages of this study, the positioning of the hole is meticulously controlled to coincide with the RCM point, ensuring the prevention of the hole's damage.

In the experiment for point cloud acquisition, the operator visually monitors the scanning area, similar to using an endoscopic camera to secure an internal view of the human body in MIS [8], [34], [35]. Once the surface of the scanned area is reconstructed, regression lines and points are created to establish the paths for palpation. In this experiment, eight regression lines are initially created at regular intervals across the entire area, followed by the addition of seven more lines between the initial eight. This method generates 6 paths from which approximately 50 points per line are extracted. Thus, this process ensures that stiffness measurements are obtained at uniformly spaced locations across the entire area.

In this experiment, the indentation depth, a crucial factor in the robotic palpation process, is set to 1 mm, in which the ball probe of the robotic palpation system has a diameter of 2 mm. Another research on estimating the stiffness of soft tissues using a ball probe validated that when the indentation depth is less than 1 mm and the model thickness is at least 9.09 mm, the stiffness of soft tissues is most accurately estimated with a 2 mm ball probe. Therefore, in this study, the indentation depth is set to 1 mm, and the model thickness is set to at least 10 mm [36].

The palpation experiments for the tumor segmentation process were conducted twice each and the quantitative results are the averages of each experiment's results. For the phantom test, we utilized both a planar phantom model and a curved phantom model, each measuring 40 mm $\times$ 40 mm. The planar phantom model included square and circular virtual tumor tissues, while the curved phantom model featured an elliptical virtual tumor tissue. These phantom models are fabricated using EcoflexTM 00-30 product in conjunction with silicone thinner. The virtual tumor tissues are formulated with a ratio of base to hardener to softener of 1:1:0, while the normal tissue is composed with a ratio of base to hardener to softener of 1:1:1.

Second, for the ex-vivo test, two distinct artificial tumor models are fabricated using porcine liver tissue with a 40 mm $\times$ 40 mm area designated as the target region. As illustrated in Fig. 6(a), one type of artificial tumor model is generated by injecting agar solution into the porcine liver tissue from a depth of 10 mm beneath the liver surface at a single location. In contrast, as depicted in Fig. 6(b), another artificial tumor model is created by injecting agar solution into the porcine liver tissue from a depth of 10 mm beneath the liver surface in a linear direction, repeating the injection process multiple times. The agar solution utilized for tumor production is prepared by boiling a mixture consisting of agar powder and deionized water in a microwave for 30 seconds, maintaining a ratio of 1:25 of agar powder to DI water [27].

III. EXPERIMENTAL RESULTS

A. Phantom Tests

Firstly, we present the outcomes of the planar phantom model test. The planar phantom model, as detailed in the Section II-E

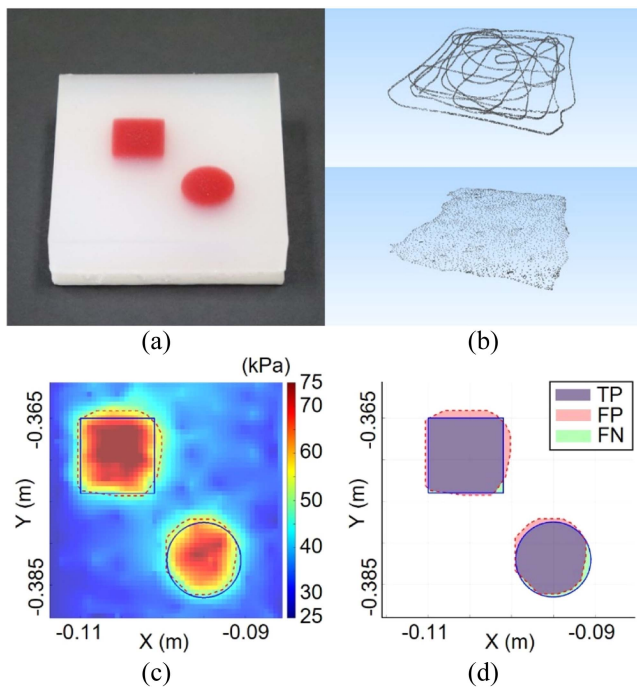


Fig. 4. Planar phantom model test, (a) planar phantom model, (b) point cloud raw data and reconstructed surface of planar phantom model, (c) stiffness map, estimated tumor outline (red dotted line) and actual tumor outline (blue solid line) of planar phantom model, and (d) sensitivity and specificity of actual tumor area and estimated tumor area (TP: True Positive, FP: False Positive, and FN: False Negative).

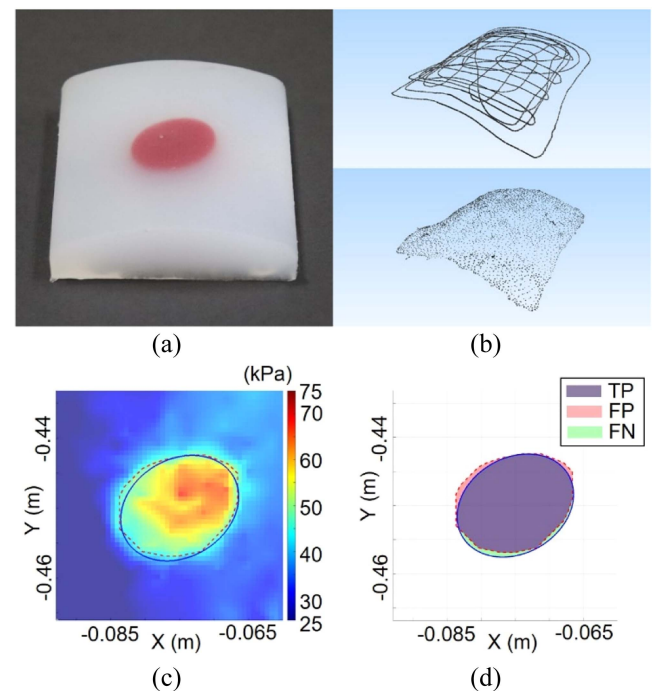


Fig. 5. Curved phantom model test, (a) curved phantom model, (b) point cloud raw data and reconstructed surface of curved phantom model, (c) stiffness map, estimated tumor outline (red dotted line) and actual tumor outline (blue solid line) of curved phantom model, and (d) sensitivity & specificity of actual tumor area and estimated tumor area (TP: True Positive, FP: False Positive, and FN: False Negative).

and depicted in Fig. 4(a), incorporates artificial tumor regions represented by a square and a circle. These tumors, highlighted in red, are fabricated with greater stiffness compared to the surrounding normal tissue. Initially, the probe moves on the model surface utilizing an image-free technique, capturing positional data at the probe tip. Subsequently, shape information is acquired in the form of a point cloud relative to the robot's base coordinate, depicted in the upper part of Fig. 4(b). During this process, the probe tip made uniform contact with the model, transmitting an average contact force of 0.068 N and applying a pressure of approximately 10.892 kPa. The raw point cloud data undergoes uniformization through the Point2Mesh deep learning algorithm, resulting in a surface characterized by uniformly distributed points, as illustrated in the lower part of Fig. 4(b). Employing the obtained surface data, a probe path is generated, which guides the probe during robotic palpation along this path. During this process, stiffness at each indentation point is calculated by employing relationships between sensed force, indentation depth, and various parameters. Post-processing and interpolation of these stiffness values enable the creation and visualization of a stiffness map, demonstrated in Fig. 4(c). Furthermore, the precise position of the artificial tumor is denoted by a solid blue line on the stiffness map and the estimated tumor boundary obtained through thresholding is indicated by a dashed red line. The stiffness map displays areas of high stiffness in red, where it's evident that the estimated regions, corresponding to high stiffness, closely align with the actual location of the

artificial tumor. In addition, a qualitative evaluation of the segmentation via thresholding of approximately 1.48 times can be conducted by assessing the ratio of the actual tumor area to the estimated tumor area. Illustrated in Fig. 4(d), areas where the actual and estimated areas intersect are defined as True Positive (TP), where the actual area intersects with the non-overlapping estimated area as False Negative (FN), where the estimated area intersects with the non-overlapping actual area as False Positive (FP), and where neither the estimated nor actual areas intersect as True Negative (TN). With these definitions, the sensitivity and specificity of the estimated tumor segmentation can be computed. In the tumor segmentation experiment utilizing the planar phantom model, the achieved values are about 0.9634 for sensitivity and about 0.9729 for specificity.

Secondly, we present the outcomes from the curved phantom model test. The curved phantom model, illustrated in Fig. 5(a), is manufactured to resemble the complex 3D shape of a real organ, thereby validating surface characteristics with a curved model distinct from the planar phantom model. As with the planar phantom model, the probe tip made uniform contact with the model, transmitting an average contact force of 0.063 N and applying a pressure of approximately 10.131 kPa. Surface data for the curved model is acquired using an image-free approach. As depicted in the upper part of Fig. 5(b), positional data from the probe tip is captured in the form of a point cloud, representing surface information. After a point2mesh learning, a uniformized surface featuring evenly distributed points, as depicted in the

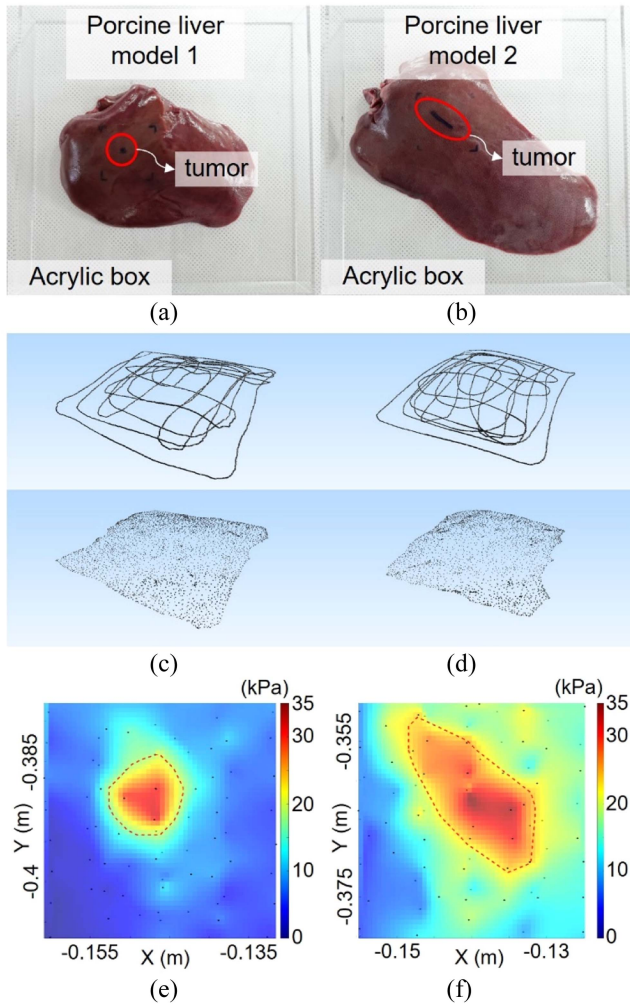


Fig. 6. Ex-vivo test, (a)–(b) porcine liver models, (c)–(d) point cloud raw data and reconstructed surfaces of porcine liver models, and (e)–(f) stiffness maps and estimated tumor outlines (red dotted line) of porcine liver models.

lower part of Fig. 5(b), is attained. Subsequently, a stiffness map is generated through robotic palpation, depicted in Fig. 5(c). Analogous to the outcomes of the planar model test, the solid blue line delineates the actual artificial tumor location, and the dashed red line indicates the estimated tumor boundary obtained through thresholding of approximately 1.85 times. Eventually, segmentation results, as displayed in Fig. 5(d), exhibit a close correspondence between actual and estimated areas, confirming highly accurate tumor segmentation with an average sensitivity of 0.9646 and an average specificity of 0.9878.

B. Ex-Vivo Tests

To ensure that the introduced method can be applied to real tissue, we conducted ex-vivo experiments using a porcine liver. Similar to the phantom test, an ex-vivo test was conducted utilizing porcine liver tumor models, applying the aforementioned three-step process. As outlined in the Section II-E, porcine liver model 1, depicted in Fig. 6(a), represents a model where a tumor

is created in a specific region of the porcine liver, while porcine liver model 2, illustrated in Fig. 6(b), represents a linearly shaped tumor on the porcine liver. Both liver models are fixed within acrylic boxes. As in the phantom experiment, in the ex-vivo experiment, the same point cloud acquisition process was followed and the position-based impedance control using the robot was well-executed for the real tissue. During the experiments, an average contact force of 0.0411 N was transmitted, and a pressure of approximately 6.536 kPa was applied. Subsequently, after learning, surface reconstruction outcomes for each model are presented in Fig. 6(c)–(d), and (e)–(f) showcase the overlay of stiffness maps generated via robotic palpation and the estimated tumor outlines derived from image-free tumor segmentation for each model. During the ex-vivo test, the stiffness contrast between the normal porcine liver tissue and the artificial tumor was discernible through the stiffness map generated by robotic palpation. This underscores the feasibility of employing an appropriate threshold of 2.86 and 1.79 times, respectively, for conducting image-free tumor segmentation on actual tissue.

IV. DISCUSSION

This paper proposes a robotic palpation system to segment tumors in Robot-assisted Minimally Invasive Surgery (RMIS) and to reduce errors in the image-matching process. Furthermore, we suggest a method of tumor segmentation that does not rely on images. To implement this, three main processes were presented: surface reconstruction process, robotic palpation process, and tumor segmentation process. Through experiments with planar and curved phantoms, we successfully generated a stiffness map, and based on this, we were able to verify that tumor segmentation is possible. Through this process, the accuracy of tumor segmentation was verified using sensitivity and specificity. The planar phantom test resulted in a sensitivity of 0.9634 and a specificity of 0.9729, while the curved phantom test resulted in a sensitivity of 0.9646 and a specificity of 0.9878. For reference, the image-based tumor segmentation method by Kirk A. Nichols et al. [2], reported a sensitivity of 0.9565 and a specificity of 0.9484. Additionally, Ranran Sun et al. [37], reported mean values of sensitivity as 0.92 ± 0.01 and mean values of specificity as 0.88 ± 0.05 . Therefore, compared to other related studies, it was confirmed that the tumor segmentation conducted in this study shows similar levels of sensitivity and specificity. Such accuracy evaluation is only possible when we know the ground truth values of the surface shape of the phantom model and the location and shape of the tumor. The model used in the phantom test was made using a 3D mold produced by a 3-dimensional printer. Because we knew the surface shape and the location and shape information of the tumor, we were able to verify the accuracy of the tumor segmentation. On the other hand, the porcine liver model used in the ex-vivo test, which was made by hardening with an agar solution using the liver extracted from a slaughtered pig, does not allow us to know the exact surface shape of the liver and the actual location and shape of the created tumor. Therefore, in the ex-vivo test, as there was no ground truth for the porcine liver model, no evaluation of accuracy was conducted. However, it was confirmed that the

shape of the segmented tumor was divided as intended during the fabrication stage. Therefore, tumor segmentation through image-free robotic palpation was successfully performed using the proposed robotic palpation system and tumor segmentation method on real soft tissue.

Also, the following considerations should be taken for clinical application. First, in clinical applications, it is important to consider how the organ we want to palpate is connected and fixed to the human body. In actual clinical settings, the tissue of the target organ can move. So, to prevent movement, a surgical towel is sometimes used to fix the organ [38]. But it is very difficult to use a device such as a surgical towel in MIS. Therefore, in order to minimize movement during contact in MIS settings, the deformation during contact must be kept at a very low level.

The deformation observed in this experiment is caused by an applied pressure of about 6.536 kPa, a contact force of 0.0411 N, and an indentation depth of 1 mm, resulting in only localized deformation of the tissue. Therefore, the overall movement of the organ can be minimized. Furthermore, with higher control performance achieved through position-based impedance control used in this study, it is possible to minimize the overall movement of the organ and reduce position errors. Therefore, the position-based impedance control used in this study needs to have good control performance. In this way, it is possible to minimize the movement of the soft tissue of the target organ and reduce the location error caused by it. In addition, reducing the depth of indentation in the robotic palpation process is a way to prevent movement and damage of the organ.

Secondly, in this study, we designed a robotic palpation system for MIS, and proposed a method of tumor segmentation that does not rely on images with this system. However, in actual clinical practice, medical imaging is already essential for diagnosing patients' diseases and planning surgery. In addition, in surgery using a robot, the registration process of matching the coordinate system of the robot, the coordinate system of the patient, and the image of the patient is inevitable. However, magnetic resonance imaging (MRI) can cause image distortion, making it difficult to accurately represent the internal structures of the human body [39]. Also, metal devices embedded in the human body or implants made of metal can distort CT and MRI images [40]. As such, there may be limitations of medical images in certain situations. In such cases, the introduction of the image-free tumor segmentation method proposed in this study should be actively considered.

Thirdly, reducing surgical time is an important issue to alleviate the burden on both patients and surgeons during the surgical process. In the case of medical imaging, it is mostly performed as preoperative preparation, whereas the three image-free processes proposed in this study are procedures performed during the surgical procedure. The execution time for point cloud acquisition is approximately 3 minutes, point2mesh deep learning takes about 2 minutes, and robotic palpation requires a minimum of 7 minutes, so the entire process takes about 12 minutes. Therefore, introducing these processes may potentially increase the surgical time. However, since the proposed processes heavily depend on the performance of the robot and controller, as well

as the learning computer used in the surface reconstruction process, it is anticipated that hardware/software improvements could significantly reduce the increase in surgical time.

Fourthly, in the proposed robotic palpation process, it is essential to assess the safety regarding as the robot's tissue contact, as the probe attached to the robot makes multiple contacts with the tissue to estimate its stiffness. Referring to another study considering damage due to pressure and indentation speed on tissues, T. Fujikawa et al. [41] reported that there was no damage to tissue capillary if the contact pressure on soft tissue is below 1.3 MPa. Additionally, it was mentioned that, when contacting the tissue with a vertical impact speed of 1.03 m/s, erythrocytes are not observed outside the capillary if the maximum contact pressure does not exceed a certain threshold. In our study, when the robot's probe indents the tissue, the indentation depth is approximately 1.0 mm, and the indentation speed is about 1.2 mm/s. Under these conditions, an average force of 0.0772 N and a maximum force of 0.1590 N were applied in indentation experiments on normal tissues. The corresponding pressures were an average of 12.281 kPa and a maximum of 25.307 kPa. It is significantly lower than the mentioned safe contact pressure value or contact speed value in the reference. Therefore, it can be considered that the proposed palpation process has a high level of safety in tissue contact. Additionally, during experiments, a safety mode was implemented in which the robot was programmed to cease operation and move away from the tissue when a force exceeding a specific threshold was detected by a 1000 Hz force/torque sensor. This measure was introduced to enhance stability and safety. However, there is a risk of tissue damage compared to non-contact cases, so it would be safer if palpation was performed by methods such as minimizing the number of contacts, reducing the indentation speed to prevent peak force on viscoelastic materials, and lowering the indentation depth.

Finally, the estimated stiffness of the normal porcine liver in this study is approximately 7–10 kPa for porcine liver model 1 and around 10–15 kPa for porcine liver model 2. When referring to other studies estimating or measuring tissue stiffness, Evren Samur et al. [42] reported the elastic modulus of porcine liver to be around 10–15 kPa for indentation depths of 2–8 mm. Blake Johnson et al. [43] reported the elastic modulus of porcine liver at strain rates of 1%/s and 25%/s to be 0.0175 and 0.0232 MPa, respectively. Additionally, F.J. Carter et al. [33] reported a mean elastic modulus of approximately 4.0 MPa for porcine liver with a thickness of 20 mm. These studies indicate different ranges of stiffness as they estimate under different conditions, and the probing methodology used in this study, may show differences in stiffness due to variations in the physical properties of the material [43]. Additionally, in this study, since the primary objective is segmentation rather than depth estimation, the tumor tissue was positioned close to the surface in the models. However, actual malignant tumor tissues may be located deeper from the surface, and their depths can vary. Although not addressed in this study, it is anticipated that creating a regression model for depth or employing depth estimation algorithms could allow for the estimation and evaluation of tumor depth.

V. CONCLUSION

This paper proposes the robotic palpation system and introduces a method for image-free tumor segmentation utilizing this system. Using the proposed robotic palpation system in the RMIS environment, the acquisition of tissue surface information via surface reconstruction, and robotic palpation guided by surface information, were conducted and tumor segmentation based on interpolated stiffness maps was performed. We focused on visualizing tissue stiffness and accurately segmenting tumors based on quantified stiffness measurements. Validation conducted through phantom model tests confirmed the high accuracy of the proposed method in tumor tissue segmentation. Furthermore, the feasibility of image-free tumor segmentation on real tissues was demonstrated via an ex-vivo test employing porcine liver models. Future endeavors will focus on enhancing the speed and precision of the image-free tumor segmentation method utilizing the proposed robotic palpation system, thereby enabling precise segmentation of more intricate and diverse tumor tissues. Additionally, it is anticipated that the proposed method will find application in various clinical scenarios beyond tumor.

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